

Blog/Audio normalization with FFmpeg

2020-08-03: Audio normalization with ffmpeg

FFmpeg is a great tool for audio/video encoding. Here we'll go through audio normalization using the loudnorm (<http://ffmpeg.org/ffmpeg-filters.html#loudnorm>) filter.

The loudnorm filter works best when you run it in two passes. The first pass checks the audio properties of the source file. Then you use those properties as input in the second pass.

1:34

Swallowtail jig, an Irish fiddle tune, played by Katy Adelson

FFmpeg option	Description
-i	Input file.
Options for the loudnorm filter:	
-filter:a loudnorm	Activates the loudnorm audio filter.
linear=true	Normalize by linearly scaling the source audio.
i=	Set integrated loudness target. Range is -70.0 - -5.0. Default value is -24.0. For EBU R128 (https://tech.ebu.ch/docs/r/r128.pdf) normalization a target of -23dB should be used.
lra=	Set loudness range target. Range is 1.0 - 20.0. Default value is 7.0.
tp=	Set maximum true peak. Range is -9.0 - +0.0. Default value is -2.0.
offset=	Set offset gain. Gain is applied before the true-peak limiter. Range is -99.0 - +99.0. Default is +0.0.
measured_I=	Measured IL of input file. Range is -99.0 - +0.0.
measured_TP=	Measured true peak of input file. Range is -99.0 - +99.0.
measured_LRA=	Measured LRA of input file. Range is 0.0 - 99.0.
measured_thresh=	Measured threshold of input file. Range is -99.0 - +0.0.
print_format=json	Enables json output which is needed to get the parameters for the second pass.
Options for the aresample filter:	
aresample=	Enable sample rate conversion.
resampler=	Selects the swr or soxr sample rate conversion engine.
out_sample_rate=	Audio samplerate.
precision=	soxr internal precision. 28-33 is considered very high quality and suitable for 24bit audio. 20 is default and is suitable for 16bit output, with noise shaping/dithering filters. Read the FAQ (http://sox.sourceforge.net/Docs/FAQ).
FFmpeg options	
-ac	Audio channels. Use 2 for stereo, 6 for 5.1 and 8 for 7.1 sources. Only needed with the aresample filter.
-ar	Audio samplerate.
-c:a	Audio codec. AAC, MP3, FLAC, WAV, etc.
-b:a	Audio bitrate.

Pass 1:

```
# ffmpeg -i audio.dts -filter:a loudnorm=print_format=json -f null NULL
```

```
[Parsed_loudnorm_0 @ 00000236cfdc12c0]
{
  "input_i" : "-23.01",
  "input_tp" : "-10.11",
  "input_lra" : "18.80",
  "input_thresh" : "-34.44",
  "output_i" : "-24.39",
  "output_tp" : "-5.99",
  "output_lra" : "10.70",
```

```
"output_thresh" : "-35.00",
"normalization_type" : "dynamic",
"target_offset" : "0.39"
}
```

Pass 2:

The loudnorm filter resamples the audio to 192KHz. So in order to resample it back down to the same as the source we use `-ar 48000` for 48KHz or `-ar 44100` for 44.1KHz.

We also want to target the overall loudness to be -23 LUFS (Loudness Units, referenced to Full Scale).

```
# ffmpeg -i audio.dts -filter:a
loudnorm=linear=true:i=-23.0:lra=7.0:tp=-2.0:offset=0.39:measured_I=-23.01:measured_tp=-10.11:measured_
LRA=18.80:measured_thresh=-34.44 -ar 48000 -c:a aac -b:a 500k audionorm.mp4
```

While the FFmpeg sample rate conversion is generally very good, there is a better sample rate converter called SoX Resampler (<http://sox.sourceforge.net/sox.html>). If your FFmpeg has `soxr` support you can enable it with the `aresample` (<http://ffmpeg.org/ffmpeg-filters.html#aresample-1>) filter - `filter:a aresample=resampler=soxr:out_sample_rate=48000:precision=28`. Combining it with the `loudnorm` filter, the full command line would be:

```
# ffmpeg -i audio.dts -filter:a
loudnorm=linear=true:i=-23.0:lra=7.0:tp=-2.0:offset=0.39:measured_I=-23.01:measured_tp=-10.11:measured_
LRA=18.80:measured_thresh=-34.44,aresample=resampler=soxr:out_sample_rate=48000:precision=28 -ac 6 -c:a
aac -b:a 500k audionorm.mp4
```

There is a tool that automates all of this. `ffmpeg-normalize` supports batch processing, copying of the video and subtitle streams, retaining the original audio and has all the options for exact control :). Head over to <https://github.com/slhck/ffmpeg-normalize/> for more details.

```
# ffmpeg-normalize *.mkv -c:a aac -b:a 500k
```

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